

Unit 7 Pre-Test Review – Graphing

MHF4U

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ANSWERS

1) Graph each of the following reciprocal functions. Start by graphing the function in the denominator. Show as much characteristic information about the graph as you can (e.g. intercepts, asymptotes with equations, other defining points, etc).

a) $f(x) = \frac{1}{x^2 - 4}$

$$y = x^2 - 4$$

$$y = (x-2)(x+2)$$

x-int: $x = -2, 2$
 x-vert: $= \frac{-2+2}{2} = 0$

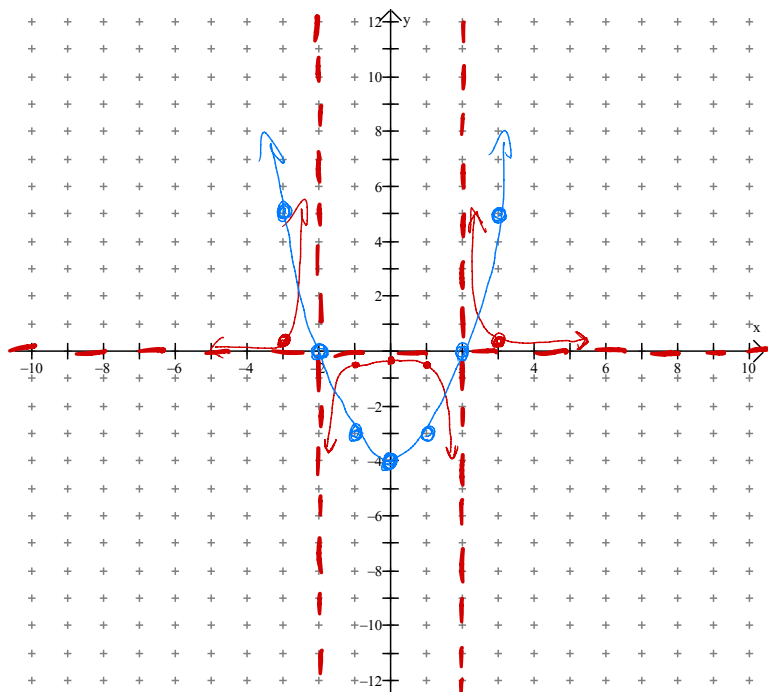
x	y
-3	5
* -2	0
-1	-3
* 0	-4
1	-3
* 2	0
3	5

$$y = \frac{1}{x^2 - 4}$$

HA: $y = 0$

VA: $x = 2$ and $x = -2$

x	$\frac{1}{y}$
-3	0.2
-2	und
-1	-0.3
0	-0.25
1	-0.3
2	und
3	0.2



b) $g(x) = \frac{2}{x+3}$

$$y = x + 3$$

x-int: $x = -3$

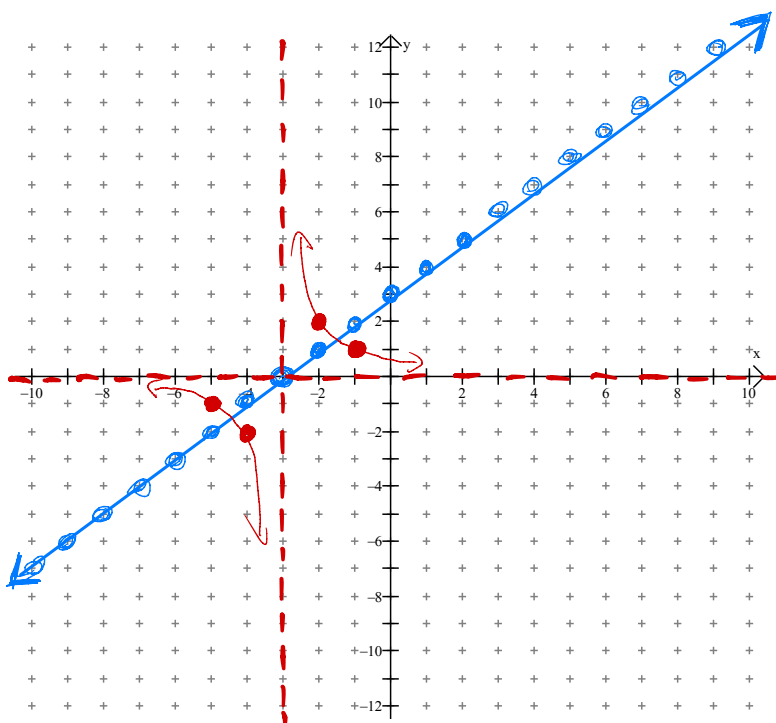
x	y
-5	-2
-4	-1
-3	0
-2	1
-1	2

$$y = \frac{2}{x+3}$$

HA: $y = 0$

VA: $x = -3$

x	$\frac{2}{y}$
-5	-1
-4	-2
-3	und
-2	2
-1	1



2) Graph each of the following functions. Make sure to identify key characteristics of the functions including intercepts and asymptotes.

a) $f(x) = \frac{x-3}{2x-1}$

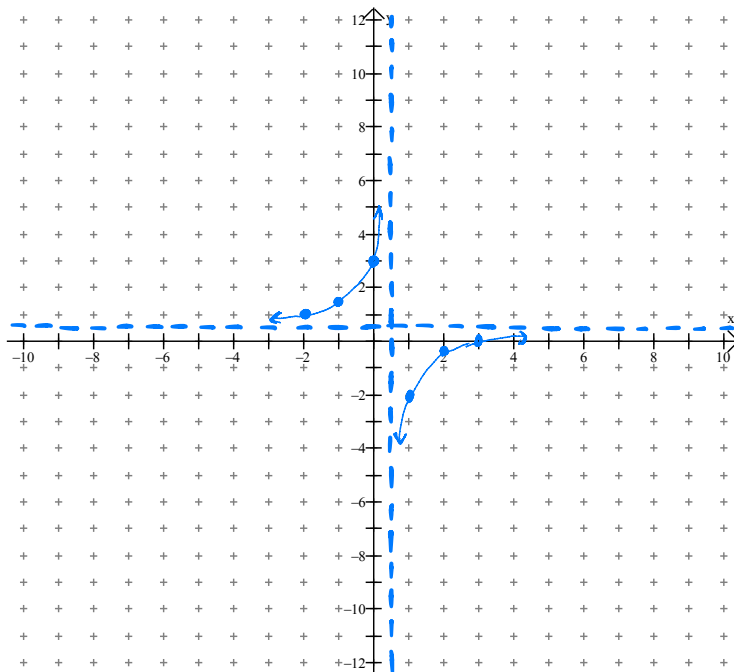
HA: $y = \frac{1}{2}$

VA: $x = \frac{1}{2}$

x-int: $0 = x-3$
 $x = 3$ (3, 0)

y-int: $\frac{0-3}{2(0)-1} = 3$ (0, 3)

other point: $f(2) = \frac{2-3}{2(2)-1} = -\frac{1}{3}$ (2, -0.33)



b) $g(x) = \frac{x+2}{x-1}$

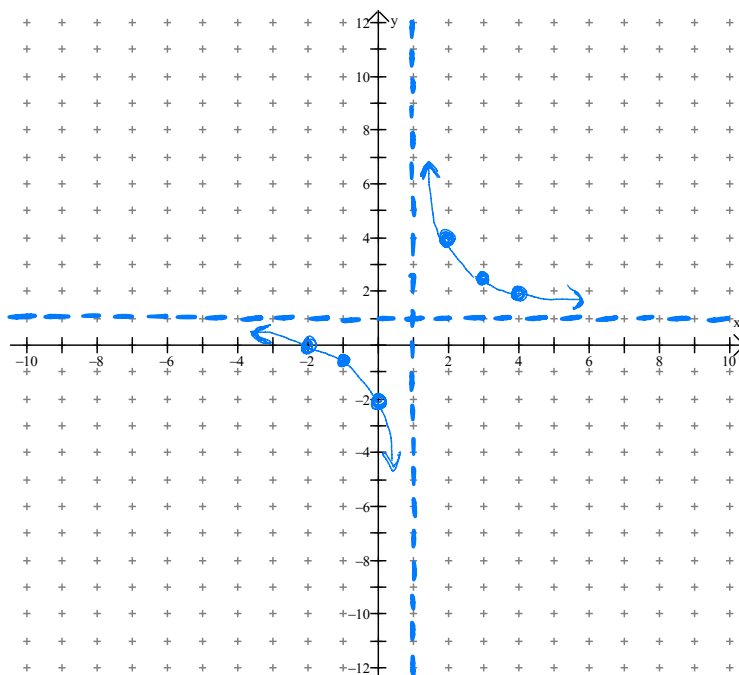
HA: $y = 1$

VA: $x = 1$

x-int: $0 = x+2$
 $x = -2$ (-2, 0)

y-int: $g(0) = \frac{0+2}{0-1} = -2$ (0, -2)

other point: $g(-1) = \frac{-1+2}{-1-1} = -\frac{1}{2}$ (-1, -0.5)



3) Use a calculator to approximate each to the nearest thousandth

a) $\ln 7$

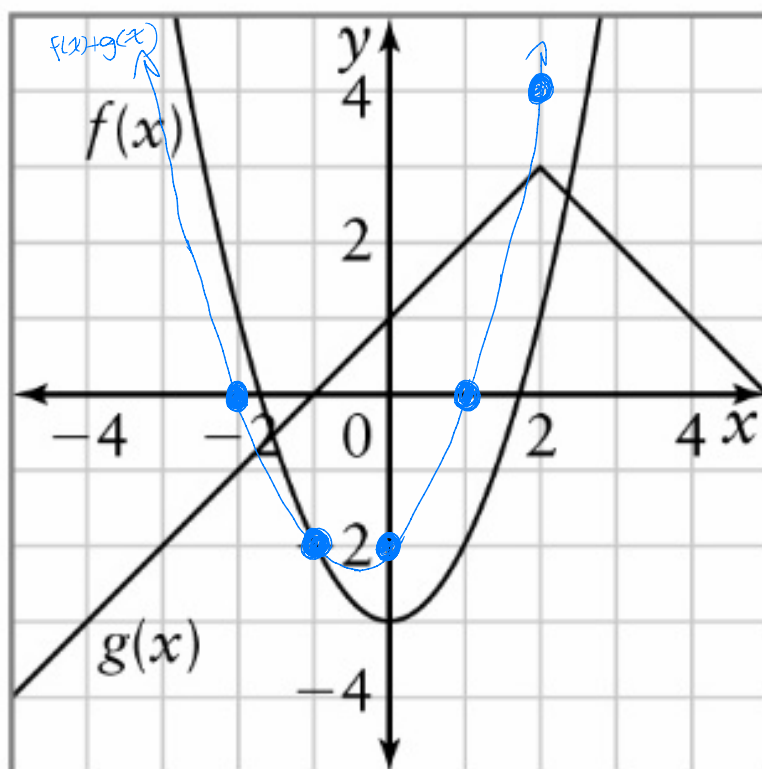
b) $\ln e$

c) e^5

4) Use the superposition principal to draw a graph of each.

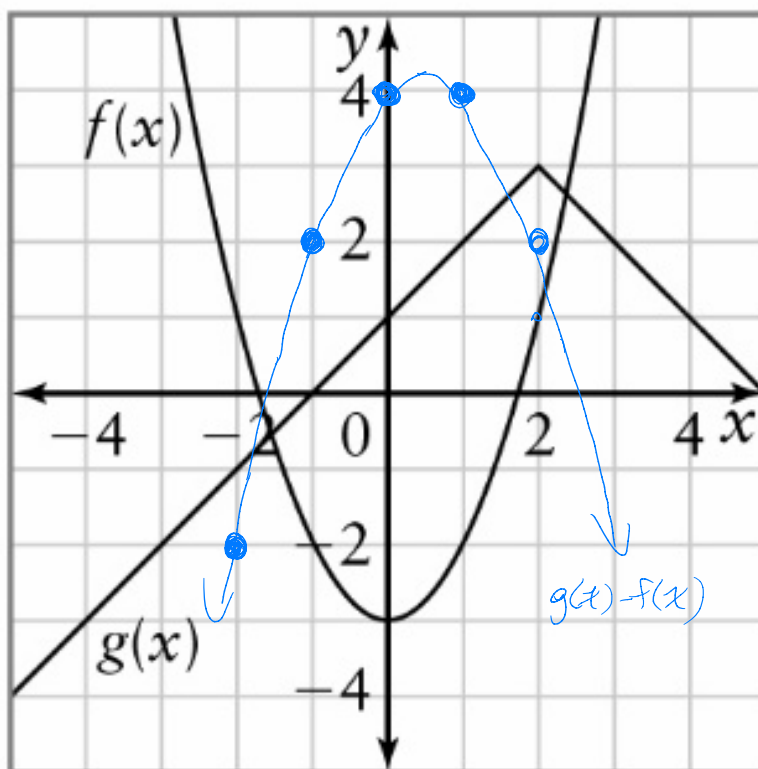
a) $y = f(x) + g(x)$

x	$f(x)$	$g(x)$	$f(x) + g(x)$
-2	1	-1	0
-1	-2	0	-2
0	-3	1	-2
1	-2	2	0
2	1	3	4



b) $y = g(x) - f(x)$

x	$g(x)$	$f(x)$	$g(x) - f(x)$
-2	-1	1	-2
-1	0	-2	2
0	1	-3	4
1	2	-2	4
2	3	1	2



5) Given that $f(x) = 2x + 3$ and $g(x) = x^2 + x$, and $h(x) = x^2 - 3$, develop an algebraic model for...

a) $y = f(x) - g(x)$

$$= 2x + 3 - (x^2 + x)$$

$$= 2x + 3 - x^2 - x$$

$$= -x^2 + x + 3$$

b) $y = f(x) - g(x) + h(x)$

$$y = -x^2 + x + 3 + x^2 - 3$$

$$= x$$

6) If $f(x) = 2x + 3$ and $g(x) = x^2 + x$, and $h(x) = x^2 - 3$, develop algebraic models for each.

a) $y = f(h(x))$

$$y = 2(x^2 - 3) + 3$$

$$y = 2x^2 - 6 + 3$$

$$y = 2x^2 - 3$$

b) $y = g(f(x))$

$$y = (2x+3)^2 + (2x+3)$$

$$y = 4x^2 + 12x + 9 + 2x + 3$$

$$y = 4x^2 + 14x + 12$$

c) $y = f(f^{-1}(x))$

$$f^{-1}(x): y = 2x + 3$$

$$x = 2y + 3$$

$$\frac{x-3}{2} = y$$

$$f^{-1}(x) = \frac{x-3}{2}$$

$$f(f^{-1}(x)) = 2\left(\frac{x-3}{2}\right) + 3$$

$$= x - 3 + 3$$

$$= x$$

7) Solve the following equations

a) $\frac{10}{x+1} = x - 2$

$$10 = (x-2)(x+1)$$

$$10 = x^2 - x - 2$$

$$0 = x^2 - x - 12$$

$$0 = (x-4)(x+3)$$

$$x = -3, 4$$

b) $\frac{x+3}{x-1} = 2x + 1$

$$x+3 = (2x+1)(x-1)$$

$$x+3 = 2x^2 - x - 1$$

$$0 = 2x^2 - 2x - 4$$

$$0 = 2(x^2 - x - 2)$$

$$0 = 2(x-2)(x+1)$$

$$x = -1, 2$$

c) $\frac{3}{3x+2} = \frac{6}{5x}$

$$3(5x) = 6(3x+2)$$

$$15x = 18x + 12$$

$$-3x = 12$$

$$x = -4$$

d) $\frac{x-2}{x+3} = \frac{x-4}{x+5}$

$$(x-2)(x+5) = (x-4)(x+3)$$

$$x^2 + 3x - 10 = x^2 - x - 12$$

$$4x = -2$$

$$x = -\frac{1}{2}$$

$$e) \frac{2}{x} + \frac{5}{3} = \frac{7}{x}$$

$$3x \left(\frac{2}{x} + \frac{5}{3} \right) = \frac{7}{\cancel{x}} (3\cancel{x})$$

$$\frac{\cancel{6x}}{\cancel{x}} + \frac{5\cancel{3}x}{\cancel{3}} = 21$$

$$6 + 5x = 21$$

$$5x = 15$$

$$x = 3$$

$$f) \frac{10}{x+3} + \frac{10}{3} = 6$$

$$3(x+3) \left(\frac{10}{x+3} + \frac{10}{3} \right) = 6(3)(x+3)$$

$$30 + 10(x+3) = 18(x+3)$$

$$30 + 10x + 30 = 18x + 54$$

$$6 = 8x$$

$$x = \frac{3}{4}$$

$$g) \frac{3}{x} + \frac{4}{x+1} = 2$$

$$x(x+1) \left(\frac{3}{x} + \frac{4}{x+1} \right) = 2(x)(x+1)$$

$$3(x+1) + 4x = 2x^2 + 2x$$

$$3x + 3 + 4x = 2x^2 + 2x$$

$$0 = 2x^2 - 5x - 3$$

$$0 = 2x^2 - 6x + 1x - 3$$

$$0 = 2x(x-3) + 1(x-3)$$

$$0 = (x-3)(2x+1)$$

$$x = -\frac{1}{2}, 3$$

$$h) 3 = \frac{6}{2x^2 - x - 4}$$

$$\frac{3(2x^2 - x - 4)}{3} = \frac{6}{3}$$

$$2x^2 - x - 4 = 2$$

$$2x^2 - x - 6 = 0$$

$$2x^2 - 4x + 3x - 6 = 0$$

$$2x(x-2) + 3(x-2) = 0$$

$$(x-2)(2x+3) = 0$$

$$x = -\frac{3}{2}, 2$$

2) Solve the following rational inequalities using a factor table

$$a) \frac{6x}{x+1} > 4$$

$$\frac{6x}{x+1} - 4 > 0$$

$$\frac{6x - 4(x+1)}{x+1} > 0$$

$$\frac{6x - 4x - 4}{x+1} > 0$$

$$\frac{2x - 4}{x+1} > 0$$

$$x\text{-int: } x = 2$$

$$\text{restrictions: } x \neq -1$$

	$-\infty$	-1	2	∞
$2x-4$	$-$	$-$	$+$	
$x+1$	$-$	$+$	$+$	
overall	$+$	$-$	$+$	

Answer:

$$x < -1, x > 2$$

OR

$$x \in (-\infty, -1) \cup (2, \infty)$$

$$b) \frac{1}{2x+10} \geq \frac{1}{x+3}$$

$$\frac{1}{2x+10} - \frac{1}{x+3} \geq 0$$

$$\frac{x+3 - (2x+10)}{(2x+10)(x+3)} \geq 0$$

$$\frac{x+3-2x-10}{(2x+10)(x+3)} \geq 0$$

$$\frac{-x-7}{(2x+10)(x+3)} \geq 0$$

$$x\text{-int: } x = -7$$

$$\text{restrictions: } x \neq -5, -3$$

	$-\infty$	-7	-5	-3	∞
$2x+10$	$-$	$-$	$+$	$+$	
$x+3$	$-$	$-$	$-$	$+$	
$-x-7$	$+$	$-$	$-$	$-$	
overall	$+$	$-$	$+$	$-$	

Answer: $x \leq -7, -5 < x < -3$

OR

$$x \in (-\infty, -7] \cup (-5, -3)$$

$$c) \frac{2x-3}{x+5} \geq \frac{2x+7}{x-3}$$

$$\frac{2x-3}{x+5} - \frac{2x+7}{x-3} \geq 0$$

$$\frac{(2x-3)(x-3) - (2x+7)(x+5)}{(x+5)(x-3)} \geq 0$$

$$\frac{2x^2 - 9x + 9 - (2x^2 + 17x + 35)}{(x+5)(x-3)} \geq 0$$

$$\frac{2x^2 - 9x + 9 - 2x^2 - 17x - 35}{(x+5)(x-3)} \geq 0$$

$$\frac{-26x - 26}{(x+5)(x-3)} \geq 0$$

$$\frac{-26(x+1)}{(x+5)(x-3)} \geq 0$$

$$x\text{-int: } x = -1$$

$$\text{restrictions: } x \neq -5, 3$$

	$-\infty$	-5	-1	3	∞
	-10	-2	0	10	
-26	-	-	-	-	
$x+1$	-	-	+	+	
$x+5$	-	+	+	+	
$x-3$	-	-	-	+	
overall	+	-	+	-	

$$\text{Answer: } x < -5, -1 \leq x < 3$$

$$\text{OR}$$

$$x \in (-\infty, -5) \cup [-1, 3)$$

$$d) \frac{7}{x-3} \geq \frac{2}{x+4}$$

$$\frac{7}{x-3} - \frac{2}{x+4} \geq 0$$

$$\frac{7(x+4) - 2(x-3)}{(x-3)(x+4)} \geq 0$$

$$\frac{7x + 28 - 2x + 6}{(x-3)(x+4)} \geq 0$$

$$\frac{5x + 34}{(x-3)(x+4)} \geq 0$$

$$x\text{-int: } x = \frac{-34}{5} = -6.8$$

$$\text{restrictions: } x \neq -4, 3$$

	$-\infty$	$-\frac{34}{5}$	-4	3	∞
	-10	-5	0	10	
$5x+34$	-	+	+	+	
$x+4$	-	-	+	+	
$x-3$	-	-	-	+	
overall	-	+	-	+	

$$\text{Answer: } -\frac{34}{5} \leq x < -4, x > 3$$

$$\text{OR}$$

$$x \in [-\frac{34}{5}, -4) \cup (3, \infty)$$

$$e) \frac{x^2 - x - 12}{x-1} < 0$$

$$\frac{(x-4)(x+3)}{x-1} < 0$$

$$x\text{-int: } x = -3, 4$$

$$\text{restrictions: } x \neq 1$$

	$-\infty$	-3	1	4	∞
	-5	0	2	10	
$x-4$	-	-	-	+	
$x+3$	-	+	+	+	
$x-1$	-	-	+	+	
overall	-	+	-	+	

$$\text{Answer: } x < -3, 1 < x < 4$$

$$\text{OR}$$

$$x \in (-\infty, -3) \cup (1, 4)$$

$$f) \frac{6x^2 - 5x + 1}{2x+1} < 0$$

$$\frac{6x^2 - 2x - 3x + 1}{2x+1} < 0$$

$$\frac{2x(3x-1) - 1(3x-1)}{2x+1} < 0$$

$$\frac{(3x-1)(2x-1)}{2x+1} < 0$$

$$x\text{-int: } x = \frac{1}{3}, \frac{1}{2}$$

$$\text{restrictions: } x \neq -\frac{1}{2}$$

	$-\infty$	$-\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{2}$	∞
	-1	0	0.4	1	
$3x-1$	-	-	+	+	
$2x-1$	-	-	-	+	
$2x+1$	-	+	+	+	
overall	-	+	-	+	

$$\text{Answer: } x < -\frac{1}{2}, \frac{1}{3} < x < \frac{1}{2}$$

$$\text{OR}$$

$$x \in (-\infty, -\frac{1}{2}) \cup (\frac{1}{3}, \frac{1}{2})$$

g) $\frac{2x-10}{x} > x-5$

$$\frac{2x-10}{x} - \frac{x-5}{1} > 0$$

$$\frac{2x-10-x(x-5)}{x} > 0$$

$$\frac{2x-10-x^2+5x}{x} > 0$$

$$\frac{-x^2+7x-10}{x} > 0$$

$$\frac{-(x^2-7x+10)}{x} > 0$$

$$\frac{-(x-2)(x-5)}{x} > 0$$

x-int: $x = 2, 5$

restrictions $x \neq 0$

	$-\infty$	0	2	5	∞
	-1	1	3	6	
-1	-	-	-	-	
$x-2$	-	-	+	+	
$x-5$	-	-	-	+	
x	-	+	+	+	
overall	+	-	+	-	

Answer: $x < 0, 2 < x < 5$
 OR
 $x \in (-\infty, 0) \cup (2, 5)$